Project Portfolio

Justus Liutala

Made with GAMMA

URBAN PLANNING | N8N

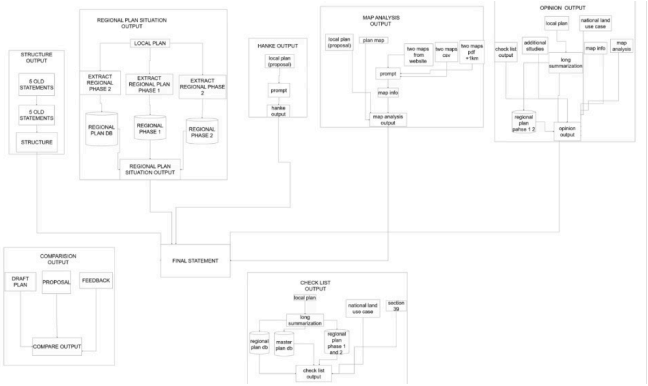
AI Handling of Land Use Documents **(RoboAI Academy)**

Project Overview

Designed and piloted an AI-supported workflow for reviewing master and detailed plans within a regional planning authority framework. The system analyzes plan documents and spatial data to assess compliance with regional plans, higher-level planning objectives, and legal requirements.

Technical Implementation

Developed checklist-based evaluation logic and map overlay analysis to support expert statements, identify planning conflicts, and detect regulatory gaps. The solution streamlines the review process while maintaining rigorous compliance standards. All in n8n and running locally



AI-Assisted Decision Support

Spatial Data Analysis

Urban & Regional Planning

Land Use Law (Finland)

GIS & Process Design

Team Role: Communication Manager

Made with GAMMA

MACHINE VISION

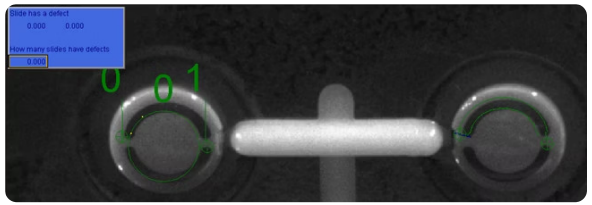
Machine Vision Quality Inspection Demo **(RoboAI Academy)**

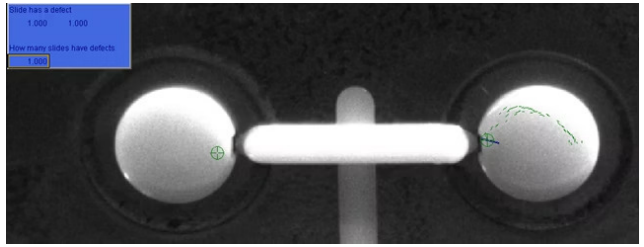
System Development

Developed machine vision system for industrial faucet quality inspection. Detects missing markings, surface defects, and line deviations.

Platform Evaluation

Evaluated and compared leading machine vision platforms (Cognex, HALCON, OpenCV) for optimal industrial inspection solutions.





1

Image Capture

High-resolution cameras acquire product images

2

Defect Detection

Algorithms identify marking inconsistencies and surface anomalies

3

Quality Assessment

Automated pass/fail decisions based on predefined tolerance thresholds

4

Data Logging

Results stored for traceability and continuous improvement analysis

Machine Vision

Advanced imaging systems for automated inspection

Image Processing

Algorithm development for defect recognition

Quality Control

Standards compliance and validation protocols

Industrial Automation

Integration with manufacturing workflows

Team Role: Project Manager

Made with GAMMA

MACHINE VISION YOLO

YOLO-Based Object Detection in Synthetic Urban Environments

This project involved the design and implementation of a deep learning-based object detection system using the YOLO architecture. The system's primary goal was to detect vehicles and pedestrians within a synthetic urban environment, which was generated using Grand Theft Auto V for realistic simulation data.

Dataset preparation was a crucial step, including image format conversion, meticulous annotation processing, and the transformation of complex 3D simulation metadata into 2D bounding boxes. This ensured compatibility with the YOLO training pipelines. Model training, validation, and evaluation were executed using Ultralytics YOLO in Google Colab, with a strong focus on dataset structuring, label accuracy, and fine-tuning training configurations for scenarios with limited data.

- Computer Vision
- Object Detection
- YOLO
- Dataset Engineering
- Synthetic Data
- Python & Deep Learning

Solo Project

Made with GAMMA

Hackathons

Hackathon 1: Robotics / Mechanical + Automation

2ND PLACE

Project Description

Awarded **2nd place** in a multidisciplinary hackathon focused on industrial automation solutions. Designed a gripper solution utilizing a vacuum-based gripping mechanism for reliably picking metal cylinders from an unsorted container.

The solution emphasized mechanical simplicity and robust object handling in industrial environments.

Key Skills Demonstrated

Mechanical Design · Robotics · Vacuum Gripping · Rapid Prototyping · Industrial Automation

Hackathon 2: AI / Computer Vision / Ethics

3RD PLACE

Project Description

Awarded **3rd place** in a hackathon focused on assistive technologies and ethical AI. Developed a proof-of-concept AI assistant using a depth camera for fall detection and elderly monitoring, with strong emphasis on privacy preservation by avoiding RGB imaging.

Demonstrated the feasibility of depth-based perception combined with AI for safe, non-intrusive monitoring in smart living environments.

Key Skills Demonstrated

AI · Computer Vision · Depth Perception · Ethical AI · Assistive Technologies · Privacy Preservation

Made with GAMMA